

Early Warning for State-Level Unemployment Shocks

Olsa Berani
Felipe Manzi
Olta Reşica

May 23, 2026

1 Introduction

Timely detection of labor market deterioration is one of the central challenges in applied macroeconomics. National recession indicators confirm downturns only after they have already begun, and for state-level policymakers this lag is particularly costly. Each state runs its own unemployment insurance system, workforce agency, and Rapid Response Team (RRT), and a national recession signal tells a state commissioner nothing about whether to deploy local resources. When that window is missed, agencies arrive after layoffs have peaked, UI systems are overwhelmed, and workers lose critical weeks of support with no advance notice. What is needed is not a coincident measure but an early warning system that signals the onset of deterioration before it appears in official statistics.

This paper builds such a system around the Sahm rule. Originally defined at the national level, the Sahm rule fires when the three-month moving average of the unemployment rate rises 0.5 percentage points above its prior twelve-month minimum, providing a simple and transparent recession signal [Sahm, 2019]. We extend the rule along two dimensions. First, spatially, by applying it to all 51 U.S. states rather than the national aggregate, generating a balanced panel of 9,180 state-month observations from 2010 to 2024. Second, temporally, by transforming it from a coincident indicator into a classifier predicting whether a shock onset will occur within the next three months. The central question becomes: not “is a shock happening?” but “will one begin within the next quarter?”

This is an instance of the hard problem in forecasting: the events we most need to predict are precisely those that occur rarely [Mueller and Rauh, 2022]. With a base rate of around 7%, positive events are sparse and standard accuracy metrics are uninformative. A classifier that always predicts “no shock” would be correct 93% of the time. This requires evaluation metrics and forecasting procedures designed for rare-event classification rather than overall accuracy. To forecast onset we combine four data sources: BLS Local Area Unemployment Statistics for the outcome variable, FRED and Department of Labor series for macroeconomic and state-level leading indicators, and Google Trends search data as a real-time text-based signal. We use these macroeconomic and financial variables because they are established predictors in the forecasting literature, especially in work on asset prices and macroeconomic forecasting, financial variables as recession leading indicators, and search behavior as a labor-market signal [Stock and Watson, 2003, Estrella and Mishkin, 1998, D’Amuri and Marcucci, 2017]. Since the goal of the project is not to provide an exhaustive review of the macro modeling literature, we use these papers as benchmark references for variable selection rather than as the object of the analysis. Search queries such as “layoffs” and “file for unemployment” carry no meaningful publication lag and tend to rise before official labor market indicators reflect the deterioration. We construct expanding-window z-scores for all Google Trends features so that each observation is normalized using only the history available at that point in time, avoiding contamination from future

events like the 2020 COVID spike. Predictions are generated using a rolling expanding-window cross-validation framework, with a gap parameter chosen to prevent label overlap given the three-month prediction horizon. This reproduces the information structure faced by a real-time forecaster. We evaluate four progressively richer model specifications and embed the results in a decision-cost framework suited to the RRT use case, where missing a shock is far more costly than raising a false alarm.

The remainder of the paper is organized as follows. Section 2 describes the data sources. Section 3 presents the methodology. Section 4 reports results. Section 5 concludes.

2 Data Sources

2.1 Outcome Variable: BLS LAUS

Our outcome data comes from the BLS Local Area Unemployment Statistics (LAUS) program. We use seasonally adjusted monthly unemployment rates for all 51 states (50 states plus DC) from January 2010 to December 2024. The panel is balanced: 51 states times 180 months gives 9,180 observations, long enough to span multiple business cycles and wide enough to support learning across states. For each state and month, we compute the Sahm gap as the three-month moving average of the unemployment rate minus its minimum over the prior twelve months. A shock is recorded when this gap first crosses 0.5 percentage points from below. We code only the first crossing month so that each episode contributes exactly one positive observation rather than a string of ones for the duration of the shock.

2.2 Text-Based Features: Google Trends

We use Google Trends as our text-based signal. Search data is attractive for this problem because it carries no meaningful publication lag and reflects household behavior before it appears in official unemployment figures. We collect monthly search indices for six terms: "unemployment," "file for unemployment," "unemployment benefits," "jobs hiring," "layoffs," and "resume." These terms capture both distress signals and job-search behavior in real time. The data were downloaded for the full 2010–2024 window in a single pull, so the 0–100 index values are comparable across time periods within each state-term series. Weekly indices are aggregated to monthly means. To avoid look-ahead bias, we apply expanding-window z-scores: at each month T , the mean and standard deviation used for standardization are computed from observations before T only. Without this step, pre-COVID z-scores would be computed using knowledge of the 2020 spike, which a forecaster in 2017 would not have had.

2.3 Macro and Leading Indicators

We merge three groups of macro and labor market series as predictive features. From FRED we take five national series: the unemployment rate (UNRATE), weekly initial jobless claims (ICSA), the VIX volatility index, the NASDAQ log-return, and the 10-year minus 2-year Treasury yield spread (T10Y2Y). National financial conditions tend to move before state labor markets do, so we include them as leading indicators. From the Department of Labor ETA-539 program we take state-level weekly initial claims, summed to monthly totals. Claims data is published faster than unemployment figures, making it one of the more timely state-level signals available. From BLS CES we take state nonfarm payroll employment, expressed as year-over-year growth. Payrolls reflect labor demand at the state level and complement the unemployment rate as a slower-moving indicator. We assume forecasts are made at the end of each month. Under this assumption, Google Trends monthly indices for month T are available to the forecaster, since Google finalizes monthly data by month end.

3 Methodology

3.1 Onset Variable Definition

We distinguish between two ways of coding a shock. Incidence codes every month a state is in a Sahm episode as a positive. Onset codes only the first month a shock begins. We use onset. This produces a substantially harder forecasting task because the model must identify transitions into deterioration rather than persistence during an already-observed episode. A model that predicts incidence is largely confirming deterioration that is already visible in the data. A model that predicts onset is trying to detect the start of a deterioration before it is reflected in official statistics, which is the operational problem the RRT faces in practice. Without an early warning system, agencies typically respond after layoffs have already peaked, UI systems are overwhelmed, and workers lose critical weeks of support. The value of the forecast is precisely in the lead time it creates.

Formally, we define onset as follows. At month T , onset is equal to 1 if the state is currently healthy, meaning no active Sahm shock, and at least one of the next three months triggers a first Sahm crossing. If the state is already in a shock at T , the onset label is set to missing rather than zero. Coding in-shock months as zeros would tell the model that producing no signal during an ongoing episode is correct behavior, which would artificially inflate specificity without improving early-warning ability.

We chose a horizon of three months because that is roughly the minimum lead time needed for a Rapid Response Team to mobilize staff, notify employers, and coordinate with other state programs before layoffs peak. A shorter horizon would give less time to act. A longer horizon would substantially increase uncertainty without providing a clear operational advantage. This is a prediction policy problem. Each month, for each state, the agency must map a predicted probability into a binary decision: activate the Rapid Response Team or wait. The cost of each error type is asymmetric and missing a shock is far more damaging than a false alarm, which means the decision threshold should not be set at 0.5 but calibrated to reflect those costs. We return to this in the decision-cost framework.

In the out-of-sample evaluation window from January 2022 to September 2024, 1,447 state-month observations have a defined onset label, of which 104 are positive. This gives a base rate of 7.2%, which is why overall accuracy is not informative for this problem and why we instead focus on metrics such as AUC-PR, sensitivity, and MCC.

3.2 Feature Engineering

We organize predictors into four groups.

Google Trends: For each of the six search terms we construct four transformations: an expanding-window z-score, a one-month difference, a three-month rolling standard deviation, and a year-over-year ratio. Each transformation targets a different aspect of the signal. The z-score captures the level of search activity relative to history. The difference captures recent momentum. The rolling standard deviation captures whether the series has become more volatile. The year-over-year ratio helps remove seasonal patterns. The expanding z-score is the most important step. Using a global mean and standard deviation would normalize, say, a 2017 observation using the full 2010–2024 distribution, which includes the 2020 COVID spike. A forecaster in 2017 would not have known about that spike. The expanding window fixes this by only using information available up to the point of the forecast.

Unemployment rate lags: We include lags of the unemployment rate at one, two, three, six, and twelve months, along with one-month and three-month changes and a three-month rolling

standard deviation. These give the model information about the recent trajectory of the labor market in each state. All lags are at least one month to respect the BLS publication schedule.

Since-variables: We include months since the last Sahm shock, cumulative shocks over the prior 24 months, and the lagged Sahm gap. These features encode where the state currently sits in the shock cycle. This is our main response to the small data problem. With around 100 shock onsets spread across 51 states and 180 months, the model does not have enough positive examples to learn cycle dynamics from raw sequences alone. Giving it the cycle position directly reduces that burden. A state that just recovered from a shock is at different risk from one that has been stable for several years, and these variables make that distinction explicit.

Macro lags: All five FRED series, state initial claims, and state payroll growth are included at a one-month lag. The yield curve spread captures the well-known predictive relationship between yield curve inversion and recessions. The VIX measures financial stress that often precedes labor market deterioration. The NASDAQ log-return reflects risk sentiment in financial markets. Initial claims are the most timely available labor market signal, published weekly rather than monthly. The national unemployment rate provides context for whether a state-level move is part of a broader national deterioration or something more idiosyncratic.

For model M4 we apply PCA to the six Google Trends z-scores, reducing them to a single component before passing them to the classifier. The PCA is fitted only on training data within each cross-validation fold so that no information from the test period influences feature construction.

3.3 Rolling Forecast Design

We use a Random Forest classifier throughout, with `class_weight="balanced"` and hyperparameters `max_depth=5`, `max_features=0.2`, and `min_samples_leaf=15`. The balanced class weight compensates for the 7% base rate by upweighting onset events during training, so the model does not simply learn to predict the majority class. The depth and leaf constraints limit tree complexity, which matters here because the number of positive training examples in any single fold is small. Forecasts are generated using an expanding-window cross-validation design: 36 splits, one test month per split, and an evaluation window running from January 2022 to September 2024. This rolling design produces time-varying forecasts that update as new data becomes available, reproducing the information structure faced by a real-time forecaster. We start in January 2022 to avoid the COVID recovery period, where the shock structure was unusual and not representative of normal labor market dynamics. Figure 1 shows the panel-wide onset base rate over time alongside three candidate out-of-sample start dates. Starting in 2021 pulls in the post-COVID recovery period, where shock dynamics were unusual and the base rate was artificially elevated. Starting in 2023 leaves only 837 observations, too few for a reliable evaluation. January 2022 gives 1,447 observations at a base rate of 7.2%, which is representative of normal labor market conditions and large enough to draw meaningful conclusions from. In-shock rows, where onset is undefined, are excluded from training automatically using `drop_na_in_y=True`.

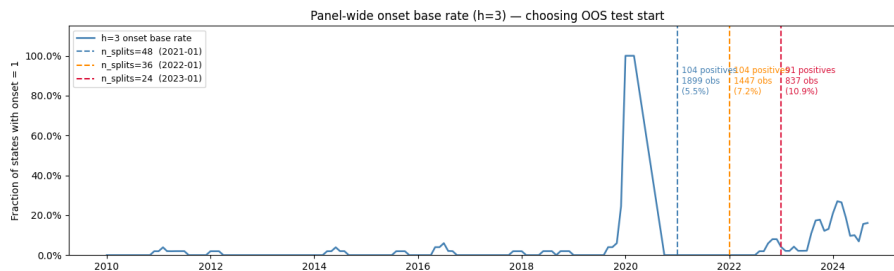


Figure 1: Panel-wide onset base rate over time.

Because the onset label is a forward variable, aggregating outcomes over the three months following the prediction date, the cross-validation design must account for this forward structure. The gap parameter is set to 2. This is necessary because our onset label at month T looks forward to $T+1$, $T+2$, and $T+3$. If training folds included months immediately before the test window, the training labels for those months would overlap with the outcomes being predicted in the test window. Setting $\text{gap} = h - 1 = 2$ eliminates this overlap and ensures that no forward information leaks from training into evaluation.

We evaluate four nested model specifications. M0 includes only the unemployment rate at lag one, plus state and month fixed effects. This is our baseline and approximates a simple persistence model. M2 adds further unemployment rate lags, since-variables, and the full set of FRED macro lags. M3 extends M2 with state-level initial claims and payroll growth. M4 extends M3 with the Google Trends features, where the six z-score inputs are first compressed via PCA before entering the classifier. The nesting is intentional: each step adds exactly one layer of information so we can isolate the marginal contribution of each feature group.

Table 1: Model specifications: feature groups included in each model.

Feature group	M0	M2	M3	M4
UR lag 1	✓	✓	✓	✓
Extended UR lags & changes		✓	✓	✓
Since-variables		✓	✓	✓
FRED macro lags (5 series)		✓	✓	✓
State initial claims			✓	✓
State payroll growth			✓	✓
Google Trends (PCA)				✓
State & month fixed effects	✓	✓	✓	✓

4 Results

4.1 Forecast Performance

Given the 7% base rate and the asymmetric costs of the problem, we prioritize sensitivity and AUC-PR over overall accuracy. A model that catches more true shock onsets, even at the cost of more false alarms, is more valuable in this setting than one that minimizes total errors.

Table 2: Out-of-sample metrics for onset forecast ($h = 3$, 2022–2024). Threshold chosen by Youden’s J . No-skill AUC-PR baseline: 0.072.

Model	AUC-ROC	AUC-PR	MCC	F1	Sensitivity	Specificity	BSS
M0	0.527	0.073	0.064	0.156	0.519	0.603	-2.73
M2	0.835	0.293	0.338	0.346	0.779	0.789	-1.64
M3	0.840	0.292	0.303	0.287	0.923	0.650	-1.54
M4	0.846	0.324	0.309	0.303	0.865	0.701	-0.99

Table 1 reports out-of-sample metrics for all four models over the evaluation window from January 2022 to September 2024, covering 1,447 observations of which 104 are positive onset events. M0 sits at AUC-PR 0.073, essentially at the no-skill baseline. M2 raises AUC-PR to 0.293 through the addition of macro controls and since-variables. M3 remains nearly unchanged at 0.292, while M4 reaches 0.324 after adding Google Trends features.

The Brier Skill Score is negative for all models, which follows from the use of balanced class weights: upweighting the minority class shifts predicted probabilities upward and inflates mean squared error for non-events. This affects calibration but not discrimination, and the ROC and PR results confirm that M2 through M4 meaningfully rank risky observations above safe ones.

At its Youden threshold, M4 correctly identifies 90 out of 104 shock onsets and misses 14. Achieving this level of recall requires accepting a large number of false positives, which is typical in rare-event problems at a 7% base rate and reflects the central tradeoff of the paper: improving early detection necessarily comes at the cost of more false alarms.

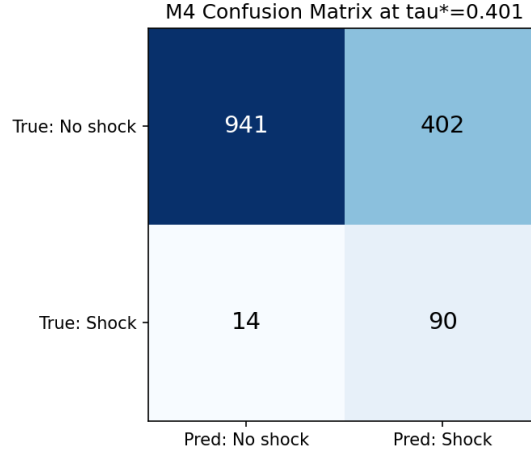


Figure 2: Confusion matrix for M4 at $\tau = 0.40$. Out of 104 shock onsets in the out-of-sample window, M4 correctly identifies 90.

4.2 ROC and Precision-Recall Curves

Figure 3 shows ROC and precision-recall curves for all four models over the out-of-sample window from January 2022 to September 2024.

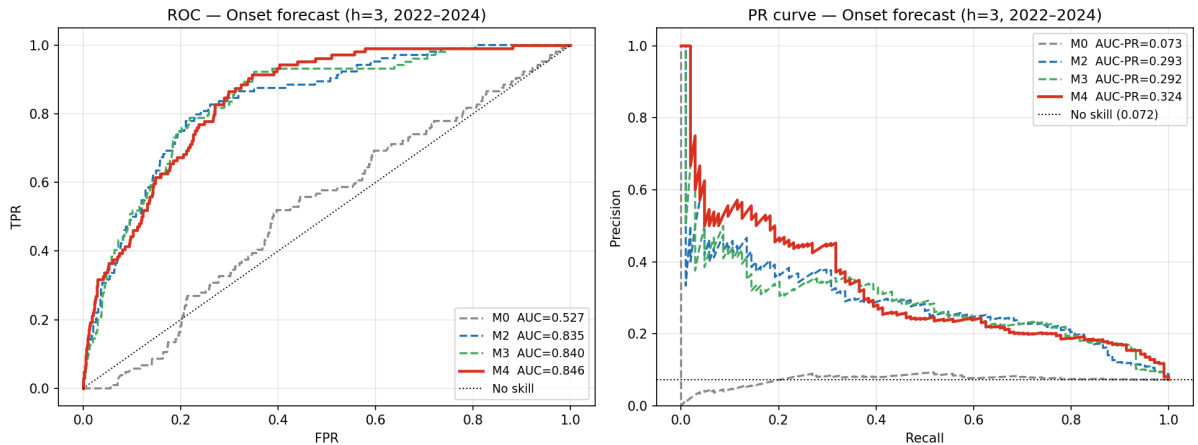


Figure 3: ROC curves (left) and precision-recall curves (right) for M0–M4, out-of-sample window January 2022 to September 2024. The dotted horizontal line in the PR plot marks the no-skill baseline at 7.2%.

With so few positive events, the ROC curve alone is not sufficient to evaluate these models. A classifier can achieve a misleadingly high AUC-ROC simply by ranking the majority class well,

without necessarily doing a good job of detecting the rare shock onsets we actually care about. The precision-recall curve is more demanding and more appropriate here. It focuses entirely on the model’s ability to retrieve shock onsets, penalizing models that produce too many false alarms relative to true detections. The no-skill AUC-PR baseline is 0.072, equal to the base rate: a model that assigns every observation the same probability would sit on this line. Any area above it represents genuine predictive content.

M0, which uses only a single unemployment rate lag, sits essentially on the no-skill line in both curves. This tells us that recent unemployment levels alone carry almost no information about whether a shock will start in the next three months. The ROC curve for M0 hugs the diagonal, and its AUC-PR of 0.073 is indistinguishable from the baseline. Persistence is not early warning. The jump from M0 to M2 is our most important result. On the ROC curve, M2 moves from near-random to an AUC of 0.835, a substantial improvement. On the PR curve the gain is even more telling: AUC-PR rises from 0.073 to 0.293, roughly four times the no-skill baseline. This means that once we give the model information about national financial conditions and where each state sits in the shock cycle, it can meaningfully distinguish pre-shock months from normal periods. The shape of the M2 PR curve is also informative: it maintains reasonable precision across a wide range of recall levels, which means the model is not just detecting easy cases but is identifying shocks that would otherwise be hard to anticipate.

M2 to M3 is essentially flat. Adding state-level claims and payroll growth raises AUC-PR by less than one percentage point. This suggests that once national conditions and cycle position are already in the model, the additional state-level series contain little incremental information. National conditions are already a good proxy for what is happening at the state level.

M3 to M4 shows a consistent improvement, with AUC-PR rising from 0.292 to 0.324. The gain from Google Trends is modest but meaningful, and its shape on the PR curve matters as much as the aggregate number. M4 diverges most clearly from M3 at high recall levels, meaning that when the threshold is lowered to catch more shocks, the Google Trends features help the model maintain better precision. At low recall the two models are similar, but as we push toward catching 80 or 90 percent of shocks, M4 produces fewer false alarms per true detection than M3. This is exactly the region that matters for the RRT decision problem. Given the asymmetric costs discussed in the onset section, the relevant operating point is at high recall rather than at the threshold that maximizes overall accuracy, and the PR curve is therefore the more useful tool for evaluating performance in the region most relevant to policymakers.

4.3 BONUS: Decision-Cost Framework

A higher AUC-PR does not automatically mean better policy outcomes. The right model depends on what errors cost the decision-maker. Here the decision-maker is a state labor commissioner choosing each month whether to activate the Rapid Response Team. We assign two costs: a false positive costs \$1M in unnecessary mobilization, and a false negative costs \$10M. The 10:1 ratio reflects the fact that a missed shock leaves workers with no advance notice and no support, while a false alarm is costly but operationally recoverable.

True positives and true negatives carry zero cost: a correct activation avoids the FN cost entirely, and correctly staying idle requires no resources. The 10:1 ratio is conservative but reasonable. A single RRT activation involves staff deployment, employer outreach, and coordination costs that are real but bounded and one-time. A missed shock affects potentially thousands of workers who receive no advance notice, lose weeks of UI support, and face employers who had no time to plan alternatives. The social cost accumulates across the full duration of the episode and is not recoverable in the way a false alarm is.

The optimal threshold τ^* is where expected total cost is minimized. On the ROC curve this is

the point where an isocost line touches the curve. Figure 4 shows that all macro-feature models choose $\tau^* < 0.5$, meaning the RRT should activate even at moderate predicted probabilities rather than waiting for near certainty before responding.

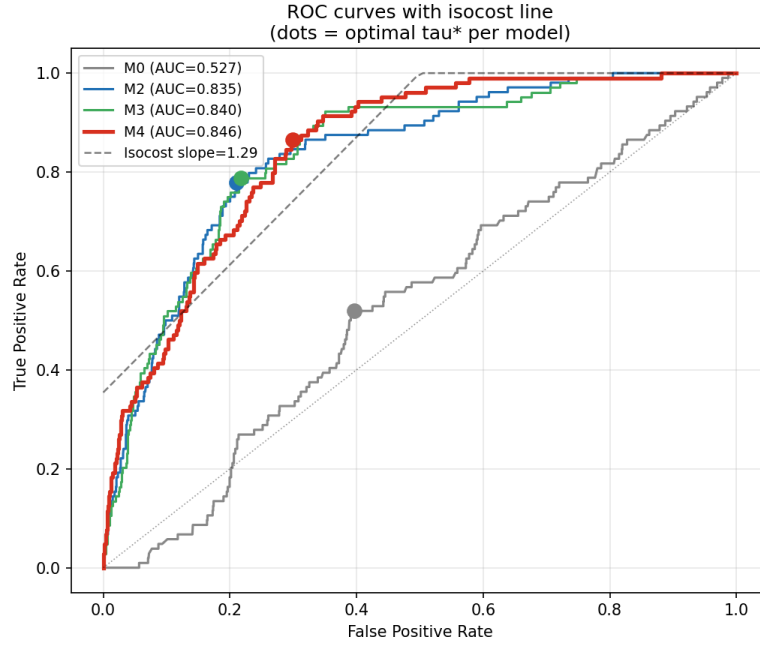


Figure 4: ROC curves with isocost line. Dots mark τ^* per model.

Figure 5 shows expected cost across the full range of thresholds. At very low thresholds the models activate too frequently, generating many false positives and steadily increasing operational cost. At high thresholds the opposite occurs: false negatives rise sharply because real shock onsets are missed. The minimum point of each curve identifies the threshold where these two types of errors are best balanced under the assumed 10:1 cost ratio.

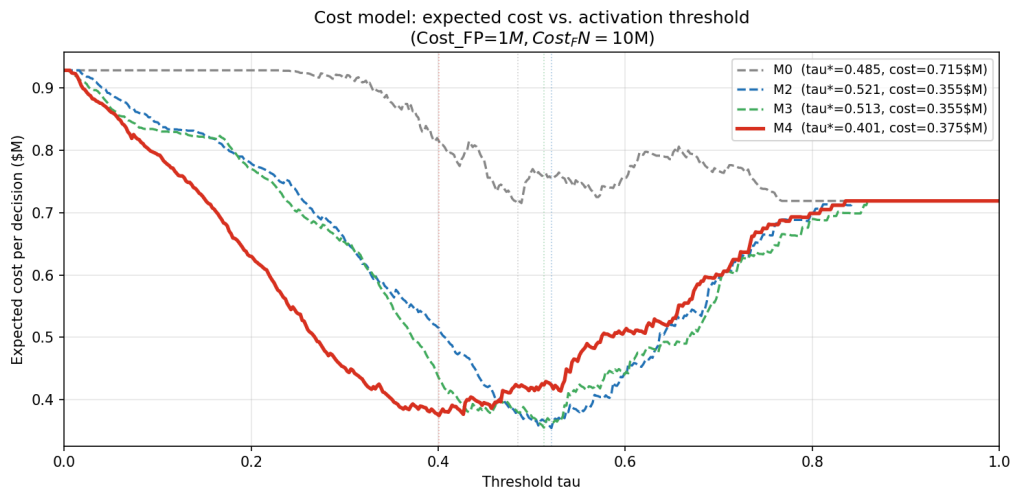


Figure 5: Expected cost per decision across thresholds. Dotted lines mark τ^* per model.

Table 3: Cost model summary at τ^* ($C_{FP} = \$1\text{M}$, $C_{FN} = \$10\text{M}$).

Model	τ^*	Cost/decision	Annual (\$M)	TP	FP	FN
M0	0.485	0.715	437.3	54	534	50
M2	0.521	0.355	217.0	81	283	23
M3	0.513	0.355	217.4	82	294	22
M4	0.401	0.375	229.2	90	402	14

M3 and M4 produce similar total costs under the 10:1 assumption, but through different tradeoffs. M4 catches 8 additional shocks compared to M3, but also generates 108 more false alarms. Figure 6 shows how the preferred model changes as the relative cost of false negatives increases. M4 is the cheaper option when the FN/FP ratio is below roughly 4:1. Above that crossover, M3 becomes preferable because its lower false alarm rate outweighs the value of catching 8 additional shocks per year.

For an agency where a missed shock means workers receive no support, M4 at $\tau \approx 0.40$ may provide the more attractive operating point despite its larger number of false alarms.

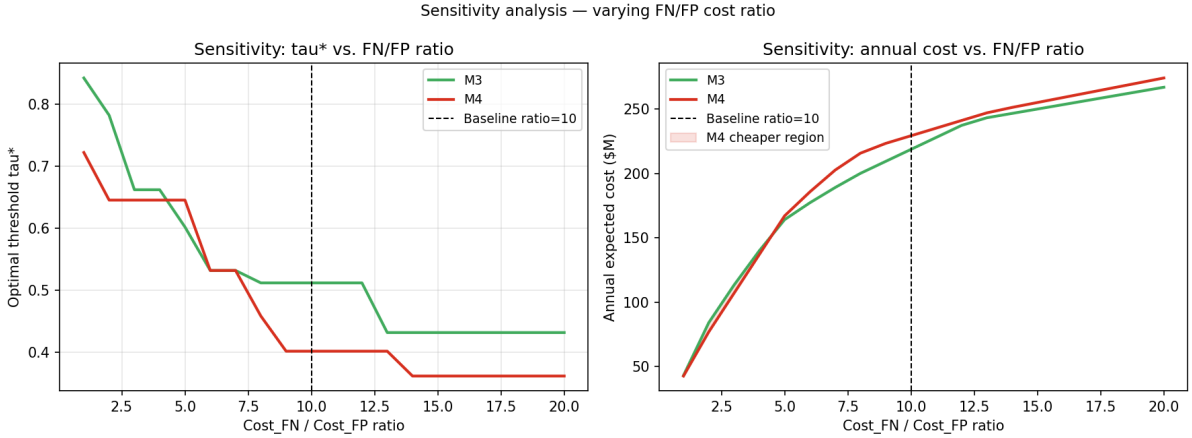


Figure 6: Sensitivity analysis over FN/FP ratios from 1:1 to 20:1. The shaded region marks where M4 is cheaper than M3.

5 Conclusion

This paper set out to build an early warning system for state-level unemployment shocks, extending the Sahm rule along two dimensions: spatially, by applying it to all 51 U.S. states, and temporally, by transforming it from a coincident indicator into a three-month-ahead onset classifier. The central challenge throughout is the hard problem: shock onsets are rare, the signal is weak, and standard accuracy metrics are uninformative at a 7% base rate. The design choices in the paper, from the onset definition to the since-variables to the expanding-window cross-validation, are all responses to this underlying difficulty.

The main finding is that knowing where a state sits in the shock cycle matters more than anything else. Adding national financial conditions and since-variables to a simple persistence model raises AUC-PR from 0.073 to 0.293. The since-variables are particularly important here: with so few positive events, the model cannot reliably learn cycle dynamics from raw sequences alone, and encoding cycle position directly reduces that burden. State-level claims and payroll data add almost nothing on top of this, suggesting that national conditions already capture the relevant

information. Google Trends adds a further consistent lift to 0.324, most visible in the high-recall region of the precision-recall curve where catching shocks matters most for the decision-maker.

The decision-cost analysis makes the practical implication concrete. A better AUC-PR does not automatically translate into lower policy cost. The right threshold depends on what errors cost the decision-maker, and in this setting missing a shock is far more costly than raising a false alarm. Under a 10:1 FN/FP ratio the RRT should activate at a threshold of around 0.40 rather than waiting for high predicted probabilities. M4 catches 8 more shocks than M3 but generates 108 more false alarms. Whether that tradeoff is worth it depends on how the agency values early detection relative to unnecessary mobilizations, but for most RRT use cases, where the cost of delayed detection is high, the more sensitive model is the better choice. The sensitivity analysis confirms this: M4 is preferred whenever false negatives are assigned more than roughly four times the cost of false positives.

The main limitation is the scarcity of events. With 104 positive observations in the evaluation window, the performance estimates carry real uncertainty and are sensitive to the choice of evaluation period. This is a structural feature of the problem rather than a failure of the method. The expanding-window rolling design is the appropriate framework for this setting, but it does not resolve the underlying scarcity of events. Future work could address it by lowering the Sahm threshold to 0.3 percentage points to generate more training examples, extending the sample further backward in time, or pooling states across countries to build a larger panel. More broadly, the results suggest that real-time search data can meaningfully complement traditional macroeconomic indicators in early warning systems. The Google Trends signal is not transformative on its own, but it adds consistent value at the margin, particularly in the high-recall region where the cost of missing a shock is highest. For policymakers facing the prediction policy problem of when to mobilize resources ahead of a labor market deterioration, a model that combines macro conditions, cycle position, and real-time search behavior offers a practical and interpretable early warning tool.

References

- Francesco D'Amuri and Juri Marcucci. The predictive power of google searches in forecasting US unemployment. *International Journal of Forecasting*, 33(4):801–816, 2017.
- Arturo Estrella and Frederic S. Mishkin. Predicting U.S. recessions: Financial variables as leading indicators. *Review of Economics and Statistics*, 80(1):45–61, 1998.
- Hannes Mueller and Christopher Rauh. The hard problem of prediction for conflict prevention. *Journal of the European Economic Association*, 2022.
- Claudia Sahm. Direct stimulus payments to individuals. In Heather Boushey, Ryan Nunn, and Jay Shambaugh, editors, *Recession Ready: Fiscal Policies to Stabilize the American Economy*. The Hamilton Project and the Washington Center for Equitable Growth, Washington, DC, 2019.
- James H. Stock and Mark W. Watson. Forecasting output and inflation: The role of asset prices. *Journal of Economic Literature*, 41(3):788–829, 2003.